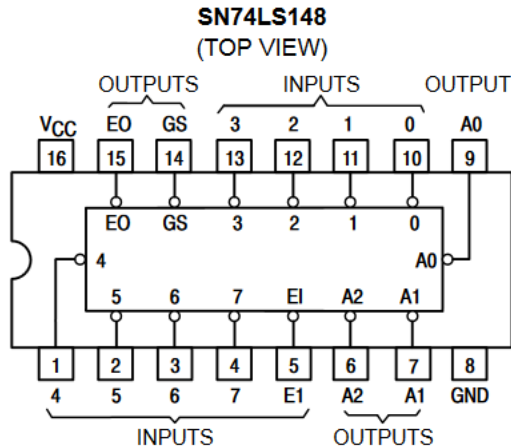
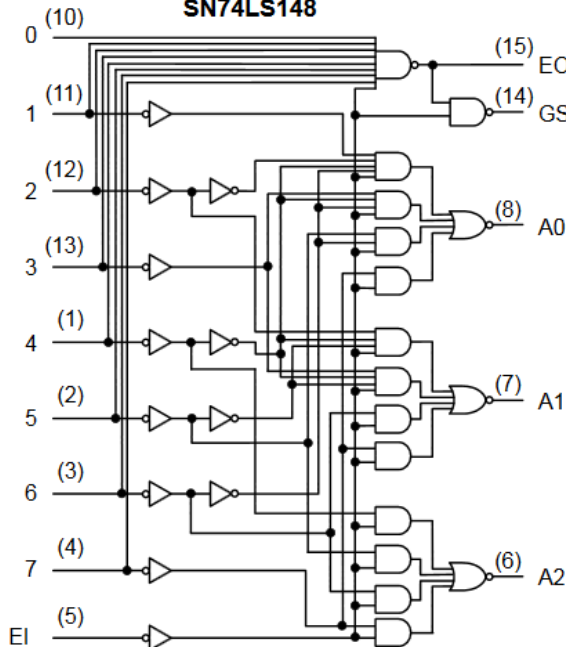


03.04.2020

<http://radio-hobby.org/uploads/datasheet/37/sn74/sn74ls147-d.pdf>**SN74LS148
FUNCTION TABLE**

INPUTS									OUTPUTS				
EI	0	1	2	3	4	5	6	7	A2	A1	A0	GS	EO
H	X	X	X	X	X	X	X	X	H	H	H	H	H
L	H	H	H	H	H	H	H	H	H	H	H	H	L
L	X	X	X	X	X	X	X	L	L	L	L	L	H
L	X	X	X	X	X	X	L	H	L	L	H	L	H
L	X	X	X	X	X	L	H	H	L	H	L	L	H
L	X	X	X	X	L	H	H	H	L	H	H	L	H
L	X	X	X	L	H	H	H	H	H	L	L	L	H
L	X	X	L	H	H	H	H	H	H	L	H	L	H
L	X	L	H	H	H	H	H	H	H	H	L	L	H
L	L	H	H	H	H	H	H	H	H	H	H	L	H

**FUNCTIONAL BLOCK DIAGRAMS
SN74LS148**

The SN74LS147 and the SN74LS148 are Priority Encoders. They provide priority decoding of the inputs to ensure that only the highest order data line is encoded. Both devices have data inputs and outputs which are active at the low logic level.

The LS147 encodes nine data lines to four-line (8-4-2-1) BCD. The implied decimal zero condition does not require an input condition because zero is encoded when all nine data lines are at a high logic level.

The LS148 encodes eight data lines to three-line (4-2-1) binary (octal). By providing cascading circuitry (Enable Input EI and Enable Output EO) octal expansion is allowed without needing external circuitry.

GUARANTEED OPERATING RANGES

Symbol	Parameter	Min	Typ	Max	Unit
V_{CC}	Supply Voltage	4.75	5.0	5.25	V
T_A	Operating Ambient Temperature Range	0	25	70	°C
I_{OH}	Output Current – High			–0.4	mA
I_{OL}	Output Current – Low			8.0	mA

AC CHARACTERISTICS ($V_{CC} = 5.0\text{ V}$, $T_A = 25^\circ\text{C}$)**SN74LS148**

Symbol	From (Input)	To (Output)	Waveform	Limits			Unit	Test Conditions
				Min	Typ	Max		
t _{PLH}	1 thru 7	A0, A1, or A2	In-phase output		14	18	ns	C _L = 15 pF, R _L = 2.0 kΩ (LS148)
t _{PHL}					15	25		
t _{PLH}	1 thru 7	A0, A1, or A2	Out-of-phase output		20	36	ns	
t _{PHL}					16	29		
t _{PLH}	0 thru 7	EO	Out-of-phase output		7.0	18	ns	
t _{PHL}					25	40		
t _{PLH}	0 thru 7	GS	In-phase output		35	55	ns	
t _{PHL}					9.0	21		
t _{PLH}	EI	A0, A1, or A2	In-phase output		16	25	ns	
t _{PHL}					12	25		
t _{PLH}	EI	GS	In-phase output		12	17	ns	
t _{PHL}					14	36		
t _{PLH}	EI	EO	In-phase output		12	21	ns	
t _{PHL}					28 30	40 45		

DC CHARACTERISTICS OVER OPERATING TEMPERATURE RANGE (unless otherwise specified)

Symbol	Parameter	Limits			Unit	Test Conditions
		Min	Typ	Max		
V_{IH}	Input HIGH Voltage	2.0			V	Guaranteed Input HIGH Voltage for All Inputs
V_{IL}	Input LOW Voltage			0.8	V	Guaranteed Input LOW Voltage for All Inputs
V_{IK}	Input Clamp Diode Voltage		-0.65	-1.5	V	$V_{CC} = \text{MIN}$, $I_{IN} = -18 \text{ mA}$
V_{OH}	Output HIGH Voltage	2.7	3.5		V	$V_{CC} = \text{MIN}$, $I_{OH} = \text{MAX}$, $V_{IN} = V_{IH}$ or V_{IL} per Truth Table
V_{OL}	Output LOW Voltage		0.25	0.4	V	$I_{OL} = 4.0 \text{ mA}$
			0.35	0.5	V	$I_{OL} = 8.0 \text{ mA}$
I_{IH}	Input HIGH Current All Others Inputs 1-7 (LS148)			20 40	μA	$V_{CC} = \text{MAX}$, $V_{IN} = 2.7 \text{ V}$
	All Others Inputs 1-7 (LS148)			0.1 0.2	mA	$V_{CC} = \text{MAX}$, $V_{IN} = 7.0 \text{ V}$
I_{IL}	Input LOW Current All Others Inputs 1-7 (LS148)			-0.4 -0.8	mA	$V_{CC} = \text{MAX}$, $V_{IN} = 0.4 \text{ V}$
I_{OS}	Short Circuit Current (Note 1)	-20		-100	mA	$V_{CC} = \text{MAX}$
I_{CCH}	Power Supply Current Output HIGH			17	mA	$V_{CC} = \text{MAX}$, All Inputs = 4.5 V
I_{CCL}	Output LOW			20	mA	$V_{CC} = \text{MAX}$, Inputs 7 & E1 = GND All Other Inputs = 4.5 V

Note 1: Not more than one output should be shorted at a time, nor for more than 1 second.